A REMARK ON OPENAI'S ODD-CHARACTERISTIC KAPLANSKY COUNTEREXAMPLE: AN EXPLICIT PRIME

[AUTHOR NAME]

Abstract. OpenAI's preprint *A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic* works over a field of order p^4, where p is the least prime factor of $(m!)^2+1$ for $m=\binom{1200}{600}$. That prime is not known. We observe that the same proof runs with the threshold-Kneser graph on the 41-subsets of [86], and with the explicit prime $p = 605425296543075877998295 3$, certified by APRCL and a Pocklington certificate. The argument is OpenAI's; we only re-select parameters and replace one estimate by an exact computation, and we list every changed line. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

1. Statement

We write [1] for the preprint `preprints/A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Odd-Characteristic-September-26-2026` of `https://github.com/openai/math`; line numbers refer to `build/sections/*.tex` at commit `08fcd61`. Theorem 1.1 of [1] reads: with $m=\binom{1200}{600}$ and p the least divisor >1 of $(m!)^2+1$, there are a field K of order p^4, a finitely generated group G with torsion and $a,b\in K[G]$ with $ab=1\neq ba$.

Theorem 1. *Let* $m=\binom{86}{41}=60542529654307587799829 60$ *and* $p = 605425296543075877998295 3$*, the least prime* $p>m$ *with* $p\equiv 1 \pmod 4$*. There are a field* K *of order* p^4*, a finitely generated group* G *containing torsion, and finite sums* $a,b\in K[G]$ *with* $ab=1$ *and* $ba\neq 1$*. The construction is that of* [1]*.*

2. The parameters that change

The proof of [1] uses m and p only through the following facts.

(F1) p is an odd prime with $p>m$, and -1 is a square mod p (Lemma op:field, `02` L7–26).
(F2) A triangle-free graph Γ on m vertices has a fitting factorization over $\mathbb{F}_p$ of size t, with $t/m<d/w$ (`02` L44–88, `05` L91–95).
(F3) $p>600$, $p\geq 5$ for the estimates in `03` L58–61, L139–146 and L207–209, which give $d/w>1/96$.
(F4) The residues $\bar{i}\in\mathbb{F}_p$ of the vertex indices are distinct (`04-gluing` L391).

For (F1) we *define* p as the least prime $>m$ with $p\equiv 1 \pmod 4$. Then $p>m$, and -1 is a square because $4\mid p-1$. This replaces the least-divisor argument at `02` L16–20. (F3) and (F4) are immediate from $p>m>600$.

For (F2) let Γ be the graph on the 41-subsets $I_1,\dots,I_m$ of [86], with $I_i\sim I_j$ iff $|I_i\cap I_j|<13$. Three pairwise adjacent sets would have union at least $3\cdot 41-3\cdot 12=87>86$, so Γ is triangle-free.

Date: October 9, 2026.

With $\alpha_i(T) = [T \subseteq I_i]$ and $\beta_j(T)$ the multilinear coefficient of $\prod_{x\in T} X_x$ in

$$Q_j(X) = \prod_{k=13}^{40} \frac{\sum_{x\in I_j} X_x - k}{41-k},$$

(the threshold-Kneser polynomial of [3, 2], as used in [1]) one gets $F_{ij} = \prod_{k=13}^{40}(|I_i\cap I_j|-k)/(41-k)$, which equals 1 on the diagonal and 0 for distinct non-neighbours ($13 \le |I_i \cap I_j| \le 40$). All denominators lie in $\{1,\dots,28\}$ and are invertible mod p. The index set is $\mathcal{T} = \{T \subseteq [86] : |T| \le 28\}$, of size $t = \sum_{j=0}^{28}\binom{86}{j} = 6211228474937453774338 4$. The estimate $t/m \le 401(5/7)^{100} < 1/100$ (`02` L79–87) is replaced by the exact comparison

$$96t = 5962779335939955623364864 < m = 6054252965430758779982760,$$

so $t/m < 1/96 < d/w$, where $d/w > 1/96$ is `03` L209. The chain $t/m < 1/100 < 1/96$ at `01` L66, `03` L4 and `05` L93 becomes $t/m < 1/96$. Nothing else in the seven sections depends on m or on the special form of p. In particular, neither vertex transitivity nor any spectral property of Γ is used. □

Remark 2 (Minimality within the mechanism). Among all threshold-Kneser instances (N, k, s) satisfying the paper's triangle-free criterion $3k - 3(s-1) > N$ and $t/m < 1/96$, $(86, 41, 13)$ has the least m. For $N \le 120$ this is checked by exhaustive search. For $N > 120$, $k - s \ge \lfloor N/3 \rfloor$ forces $m > 96t \ge 96\binom{121}{40} > 10^{34}$. Keeping the constant $1/100$ instead, $(86, 42, 14)$ works, with $p = m + 313 = 64866996058186701214104 13$. A sharper version of `03` L137–154 (true bad fraction $\approx 1/(48p)$ instead of $1/2$) would allow $(74, 35, 11)$; we do not claim this.

3. Certificates

- Primality: $p-1 = 2^3\cdot 3\cdot 19\cdot 60101\cdot 220909311629180017$, each factor APRCL-prime. Pocklington witnesses a for the factors $r = 2, 3, 19, 60101, 220909311629180017$ are $5, 2, 2, 2, 2$: $a^{p-1} \equiv 1$ and $\gcd(a^{(p-1)/r} - 1, p) = 1$. PARI/GP's `isprime(p,2)` (APRCL) also returns 1.
- $p \nmid (m!)^2 + 1$ (Wilson: $m! \equiv -1/192!$), so the original definition of p cannot simply be kept.
- Fit identities: exact rational check of $F(x) = \prod_{k=13}^{40}(x-k)/(41-k)$ for every $x \in \{0,\dots,41\}$ via the Stirling-number form of the multilinear reduction; plus a full symbolic expansion on the toy instance $(10, 4, 2)$ (210 vertices, 0 errors).

References

[1] OpenAI, *A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic*, OpenAI Math Release preprint, September 26, 2026, `https://github.com/openai/math/tree/main/preprints/A-Counterexample-to-Kaplanskys-Direct-Finiteness-Conjecture-in-Odd-Characteristic-September-26-2026`.

[2] A. Golovnev and I. Haviv, *The (Generalized) Orthogonality Dimension of (Generalized) Kneser Graphs: Bounds and Applications*, Theory of Computing 18 (2022), no. 22, 1–22.

[3] I. Haviv, *On Minrank and the Lovász Theta Function*, APPROX/RANDOM 2018, LIPIcs 116, 13:1–13:15.